Piecewise-Defined Functions

Name _____

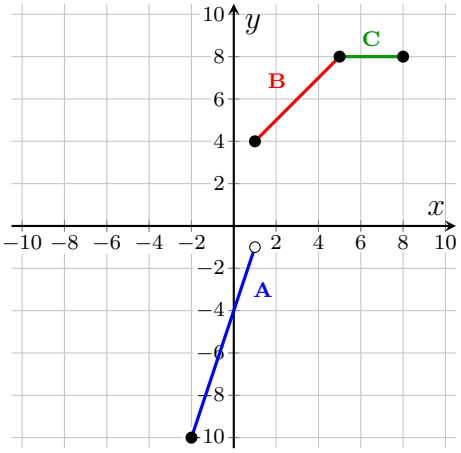
Algebra 2 Lesson 1-3 (fast track) Date _____ Period _____

Learning goal: I can graph a piecewise-defined function and write its rule from a graph.
Language goal: I can describe a graph using the words *interval*, *piece*, *slope*, and *open point / closed point* in complete sentences.

On the quiz: You will (a) read a piecewise rule from a graph and write one yourself (one piece per interval, with the right $<$ or $\leq$), and (b) graph a step function from a table and state its domain and range.

On the Topic 1 test: Question 6 (graph a three-piece function with open and closed endpoints) comes from Example 2, which this packet skips.

Example 3: Write a piecewise-defined rule from a graph



Three pieces, so three intervals. Each piece is a line, so find a slope and one point, then solve for b.

Piece A	**Piece B**	**Piece C**
interval $-2 \leq x < 1$	interval $1 \leq x \leq 5$	interval $5 < x \leq 8$
point $(-2, -10)$, $m = 3$	point $(1, 4)$, $m = 1$	point $(5, 8)$, $m = 0$
$y = mx + b$	$y = mx + b$	$y = mx + b$
$-10 = 3(-2) + b$	$4 = 1(1) + b$	$8 = 0(5) + b$
$b = -4$	$b = 3$	$b = 8$
$f(x) = 3x - 4$	$f(x) = x + 3$	$f(x) = 8$

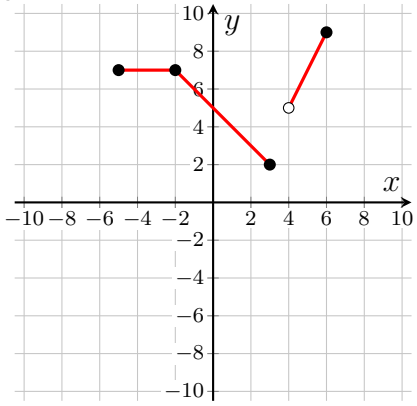
The rule is $f(x) = \begin{cases} 3x - 4, & -2 \leq x < 1 \\ x + 3, & 1 \leq x \leq 5 \\ 8, & 5 < x \leq 8 \end{cases}$

Why $x < 1$ and not $x \leq 1$ for Piece A? The point $(1, -1)$ is **open**; $x = 1$ belongs to Piece B, where the point $(1, 4)$ is **closed**. Each x gets exactly one piece.

Say it: Piece A is the line with slope _____ on the interval _____. Piece B is the line with slope _____ on the interval _____. Piece C is _____.

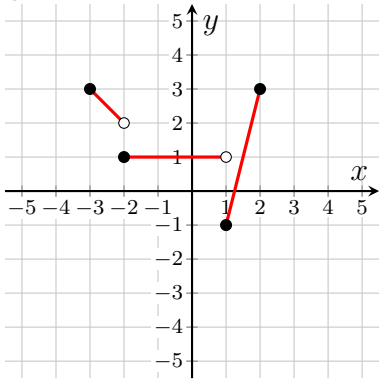
Try It 3: Write the rule for each graph

a.



$f(x) = \begin{cases} _____, & _____ \\ _____, & _____ \\ _____, & _____ \end{cases}$

b.



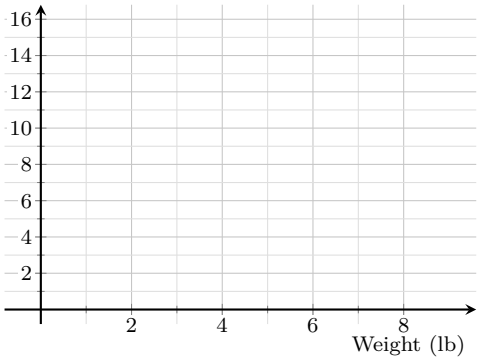
$f(x) = \begin{cases} _____, & _____ \\ _____, & _____ \\ _____, & _____ \end{cases}$

Example 5: Graph a step function

The shipping cost y of an online order depends on the weight x of the items. Graph the function, then state the domain and range and what they represent. Each y-value goes with a whole **interval** of x-values, so each piece is a horizontal segment, **open** on the left and **closed** on the right. The graph looks like a staircase: a piecewise function made only of horizontal pieces is a **step function**.

Weight of items (lb)	$0 < x \leq 2$	$2 < x \leq 4$	$4 < x \leq 6$	$6 < x \leq 8$
Shipping cost ($)	5	8	11	14

Shipping cost ($)



Domain: $\{x \mid _____ < x \leq _____\}$

The domain represents:

Range: {_____, _____, _____, _____}

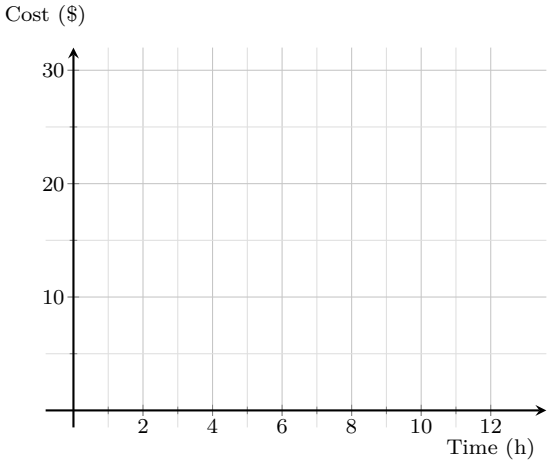
The range represents:

Minimum: _____ Maximum: _____

Try It 5: Graph a step function

The table gives the fees for a parking lot. Graph the function. State the domain and range, what they represent, and the minimum and maximum values.

Time (h)	$0 < t \leq 3$	$3 < t \leq 6$	$6 < t \leq 9$	$9 < t \leq 12$
Cost ($)	10	15	20	25



Domain: $\{t \mid$ _____ $< t \leq$ _____$\}$

The domain represents:

_____

Range: {_____, _____, _____, _____}

The range represents:

_____

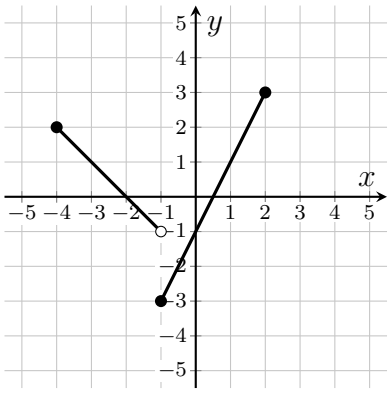
Minimum: _____ Maximum: _____

IXL: Practice for the quiz

Code	**Skill**	**Feeds**
RVD ★	Graph piecewise-defined functions	Quiz questions 1, 3, 4
W67 ★	Find the slope of a linear function	Quiz 1 and 3: the slope of each piece
WRZ	Evaluate piecewise-defined functions	Warm-up, optional

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Write the rule



$$f(x) = \begin{cases} _____, & _____ \\ _____, & _____ \end{cases}$$

Write the rule for this function. For each piece: name the interval (check the open and closed points), find the slope, use a point to find b.

Piece 1: interval _____ slope _____

_____

_____

Piece 2: interval _____ slope _____

_____

_____